

## LESSON 5-5 • SESSION 2 • PRACTICE • FORM B

## Missing Fractional Edges

Learning target – I can find a missing edge of a prism and find volumes with fractional edge lengths. · 6.G.A.2

## REMEMBER

- $V = l \times w \times h$  works with fractions:  $12 \times 10 \times \frac{3}{4} = 120 \times \frac{3}{4} = 90$ .
- $1\frac{1}{2}$  of a number is the number **plus half** of it.
- **Missing edge:** multiply the two known edges, then **divide** the volume by that product.

## WORKED EXAMPLE

A box holds  $72 \text{ cm}^3$ . It is 6 cm long and 4 cm wide.**Step 1** Known edges:  $6 \times 4 = 24 \text{ cm}^2$ .**Step 2** Divide:  $72 \div 24 = 3$ .**Step 3** Check:  $6 \times 4 \times 3 = 72$ .**Answer height = 3 cm****WORDS** edge (arista) · fraction (fracción) · volume (volumen)

## PRACTICE

show your work · label units

1. Find each volume or missing edge.

a.  $\frac{1}{2} \text{ m} \times \frac{1}{2} \text{ m} \times 6 \text{ m} \text{ } \underline{\hspace{1cm}} \text{ m}^3$

b.  $12 \text{ in} \times 9 \text{ in} \times \frac{1}{2} \text{ in} \text{ } \underline{\hspace{1cm}} \text{ in}^3$

c.  $V = 240 \text{ in}^3$ ,  $w = 6 \text{ in}$ ,  $h = 5 \text{ in}$ ;  $l = \underline{\hspace{1cm}} \text{ in}$

d.  $V = 105 \text{ in}^3$ ,  $B = 15 \text{ in}^2$ ;  $h = \underline{\hspace{1cm}} \text{ in}$

- 2.
- Shipping Box Limit.**
- A shipping box holds
- $1,080 \text{ in}^3$
- . It is 8 in wide and 9 in tall. Boxes may be at most 13 in long.

**Hint** Multiply the two known edges, then divide.

a. Multiply the two known edges.  $\underline{\hspace{2cm}} \text{ in}^2$

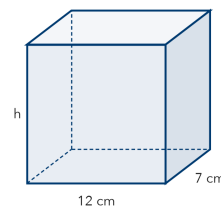
b. Find the length.  $\underline{\hspace{2cm}} \text{ in}$

c. Does the box meet the rule?  $\underline{\hspace{2cm}}$

Show your work.

- 3.
- Planter Depth.**
- A planter holds
- $1,008 \text{ cm}^3$
- of soil. Its base is 12 cm by 7 cm. How tall is it?

Show your work.

**Answer** \_\_\_\_\_ cm

- 4.
- Thin Box.**
- A flat box is 24 in long, 12 in wide, and
- $\frac{3}{4}$
- in tall. Find its volume.

**Hint** Find the base, then take  $\frac{3}{4}$  of it.

Show your work.

**Answer** \_\_\_\_\_  $\text{in}^3$ 

- 5.
- Seed Flat.**
- A seed flat is 18 in long, 10 in wide, and
- $1\frac{1}{2}$
- in deep. Find its volume.

**Hint**  $1\frac{1}{2}$  means the base plus half the base.

Show your work.

**Answer** \_\_\_\_\_  $\text{in}^3$

6. **Concrete Slab.** A concrete slab is 20 ft long, 8 ft wide, and  $\frac{1}{2}$  ft thick. Concrete costs \$6 per  $\text{ft}^3$ . Find the volume and the cost.

**Hint** The thickness is the height.

Show your work.

**Volume, cost** \_\_\_\_\_

7. **Aquarium Width.** An aquarium holds  $4,500 \text{ in}^3$  of water. It is 25 in long and 20 in tall. The width label wore off. Find the width.

Show your work.

**Answer** \_\_\_\_\_ in

8. **Planter Box.** A planter is  $\frac{3}{4}$  m long,  $\frac{1}{2}$  m wide, and 4 m tall. Soil costs \$20 per  $\text{m}^3$ .

- a. Find the volume. \_\_\_\_\_  $\text{m}^3$   
 b. Find the cost to fill it. \_\_\_\_\_  
 c. A second planter is twice as tall. Find its volume. \_\_\_\_\_  $\text{m}^3$

Show your work.

### FIND THE ERROR

explain the mistake, then fix it

9. **Jonah's Work.** A box holds  $2,520 \text{ cm}^3$ . It is 12 cm wide and 10 cm tall. Jonah wrote: " $2,520 \div 12 = 210$ , so the length is 210 cm." Explain the mistake and find the length.

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### TEST PRACTICE

choose the best answer

10. A prism has a volume of  $96 \text{ in}^3$  and a base area of  $12 \text{ in}^2$ . Its height is –  
**A** 8 in                      **B** 84 in                      **C** 108 in                      **D** 1,152 in
11. A block is  $\frac{1}{2}$  ft by  $\frac{1}{2}$  ft by 8 ft. Its volume is –  
**A**  $2 \text{ ft}^3$                       **B**  $4 \text{ ft}^3$                       **C**  $8 \text{ ft}^3$                       **D**  $9 \text{ ft}^3$

### SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

|                       |                          |                        |                      |                        |                      |                        |                        |                       |                       |
|-----------------------|--------------------------|------------------------|----------------------|------------------------|----------------------|------------------------|------------------------|-----------------------|-----------------------|
| <b>15</b><br><b>M</b> | <b>1 1/2</b><br><b>L</b> | <b>480</b><br><b>I</b> | <b>3</b><br><b>Y</b> | <b>180</b><br><b>Z</b> | <b>9</b><br><b>P</b> | <b>216</b><br><b>L</b> | <b>270</b><br><b>T</b> | <b>84</b><br><b>Q</b> | <b>12</b><br><b>U</b> |
| #2b                   |                          | #3                     | #4                   | #5                     | #6                   | #7                     | #8a                    | #8c                   |                       |

### CHECK YOURSELF

☐ I can do this on my own    ☐ I need a little help    ☐ I have a question about #\_\_\_\_\_